Fiscal Year:	FY 2019	**Task Last Updated:**	FY 07/08/2025
PI Name:	Goukassian, David A M.D., Ph.D.		
Project Title:	Space Flight Associated Changes in Astronauts' Plasma Derived miRNA Expression: Biomarker Identification		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	TRISH--TRISH		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:	NOTE: PI moved to Icahn School of Medicine at Mount Sinai from Temple University in October 2018.		
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	TRISH--Focused Investigations
Start Date:	05/01/2018	**End Date:**	07/31/2021
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	3	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	TRISH
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:	Pre/Post Flight		
Flight Assignment:	Shuttle specimen work NOTE: End date changed to 7/31/2021 per E. Urquieta/TRISH (Ed., 4/28/21) NOTE: End date changed to 10/31/2019 per E. Urquieta/TRISH (Ed., 5/23/19)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Kishore, Raj Ph.D. (Temple University) Coleman, Matthew Ph.D. (Institute for Advanced Study)		
Grant/Contract No.:	NNX16AO69A-FIP0005		
Performance Goal No.:			
Performance Goal Text:			

Task Description:

Update (Ed., 7/8/25):
Space travel-associated stressors such as microgravity or radiation exposure have been reported in astronauts after short- and long-duration missions onboard the International Space Station (ISS), with adverse effects ranging from muscle atrophy, impaired immune response, and adverse cardiovascular events. Space flight may induce remarkable miRNA expression changes in bodily fluid {i.e., peripheral blood plasma) exosomes that may reflect alterations in multiple genes and protein pathways in various tissues and cells. Additionally, a significant number of studies suggest that differential levels of circulating cell-free mtDNA (cf-mtDNA) play a significant role in various human diseases, such as diabetes, rheumatoid arthritis, neurodegenerative, and CV diseases. This may become an important test to identify, model, and validate predictive biomarkers in blood and other bio-fluids to allow for sub-clinical detection of disease development and serve as a reference for the human sample database. Overall, our proposed work will lay the foundation for developing mitigating factors to prevent dysfunction in astronauts during and after extended deep space missions.

Focused Investigation Project

This project aims to evaluate the potential impact of the space flight environment on the regulation of molecular pathways mediating cellular stress responses. We performed a first-of-its-kind pilot feasibility study to assess space flight-associated changes in exosomes derived from peripheral blood (PB) plasma collected 10 days before the launch (L -10) and the day of landing (R 0) from two astronauts who participated in STS-100 and STS-104 missions. Our preliminary pilot experiment results suggest that spaceflights may induce remarkable changes in the cargo of circulating plasma exosomes that may reflect alterations in multiple gene and protein pathways in various tissues and cells. This work represents a pilot/feasibility study to identify plasma exosomal miRNA as a source of blood derived epigenetic biomarker identification.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

Task Progress:

Study I: Biomarker development using astronaut peripheral blood exosomes.
Hypothesis and Specific Aims: Our central hypothesis is that space travel-associated stressors (radiation, microgravity, etc.) induce modifications in the exosomal miRNA contents, and their delivery to target cells/tissues can contribute to disease pathogenesis. Thus, these modifications can be used as preclinical prognostic biomarkers for health and disease.

Specific Aim 1: Examine miRNA exosomal cargo modifications in the blood plasma exosomes from 14 astronauts.

Specific Aim 2: Perform in vitro validation of human cell responses to exosomal miRNA alterations.

Deliverables: Blood samples were collected at three designated time points (at 10 days before the launch (L-10), within 2-3 hours after landing (R-0), and 3 days after landing (R+3)) from each of the 14 astronauts and processed for miRNA profiling using next-generation sequencing (NGS). Using state-of-the-art bioinformatics tools, we determined the predictive/prognostic biomarker value of space travel-associated modifications in plasma samples of 14 astronauts. We utilized various miRNA, lncRNA and their target mRNA databases and protein-protein network interactions, including currently available space travel-associated OMICs databases across all species.

Our studies determined that: (i) stress conditions in space environment induce modifications in the exosomal miRNA, lncRNA contents; (ii) modifications in exosomal cargo in astronauts' blood can be used as preclinical prognostic biomarkers for health and disease.

Publications in preparation: 1) Astronaut exosomal miRNA profiles; 2) In vitro validation of exosomal miRNA alterations; 3) Exosomal long non-coding RNAs.

Study II: Biomarker development using cf-mtDNA and peripheral blood mononuclear cells (PBMCs).

Hypothesis and Specific Aims: Our central hypothesis is that space travel-associated stressors induce differential levels of circulating cell-free mtDNA which play a significant role in various human diseases and may serve as biomarkers of cellular stress.

Specific Aim 1: Examine space flight effects on circulating cf-mtDNA in astronaut peripheral blood.

Specific Aim 2: Assess corresponding stress and immune responses in PBMCs from astronauts.

Deliverables: Blood samples collected at three designated time points (at 10 days before the launch (L-10), within 2-3 hours after landing (R-0), and 3 days after landing (R+3)) from 14 astronauts were processed to measure the abundance of cf-mtDNA compared to nuclear DNA by real-time quantitative PCR using primers for MT-CO1 and MT-CO3. Peripheral blood extracellular vesicles (EVs) were isolated from 3 astronauts and quantify EV-mtDNA via qPCR; purified exosomal RNA were analyzed by small RNA sequencing as well. PBMCs were isolated from blood samples collected at L-10, R-0, R+3 from 6 astronauts to measure the expression of genes encoding inflammation, oxidative stress, and DNA damage markers, as well as oxidative damage of DNA/RNA.

Our studies determined that: (i) cf-mtDNA abundance might serve as a potential biomarker of stress or immune response related to stress conditions or environmental factors associated with space flight.

Overall Impact: The implications of these studies may have immediate operational relevance as in the short period of time large number of blood and other bodily fluid samples collected retrospectively from astronauts that flew earlier missions could be analyzed. The very stable nature of exosomes may allow for retrospective analyses of a large number of samples obtained during and after space travel. Any epigenetic retrospective OMICs findings in the samples collected over the last 10-15 years could be used as reference databases for prospectively collected samples. Both could be correlated with the health of astronauts by medical professionals involved in monitoring astronauts' health during and after space flights. To this extend, we currently work with NASA's Life Sciences Data Archive (LSDA), Lifetime Surveillance of Astronaut Health (LSAH), and the NASA Institutional Review Board (IRB) to obtain retrospective clinical data for the 14 astronauts to validate/substantiate the predictive clinical value of our findings. Nevertheless, this feasibility study underscores the need for a comprehensive study with a large enough sample size that will allow having reliable statistical power to suggest predictive outcomes for preclinical prognostic biomarkers.

Location of the Studies: The Cardiovascular Research Center at the Temple University School of Medicine and the

	Cardiovascular Research Institute at the Icahn School of Medicine at Mount Sinai. Timeline: Quarter 1 and 2 - Exosomal miRNA Profiling (NextGen Sequencing) and Validation; Quarter 2 and 3 - Bioinformatics and Statistical Analyses; Quarter 3 and 4 - Biomarker Identification, Data Deposition and Publications(s).
Bibliography Type:	Description: (Last Updated: 04/23/2026)
Articles in Peer-reviewed Journals	Brojakowska A, Kour A, Thel MC, Park E, Bisserier M, Garikipati VNS, Hadri L, Mills PJ, Walsh K, Goukassian DA. "Retrospective analysis of somatic mutations and clonal hematopoiesis in astronauts." Commun Biol. 2022 Aug 17;5:828. https://doi.org/10.1038/s42003-022-03777-z ; PMID: 35978153; PMCID: PMC9385668 , Aug-2022
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Articles in Peer-reviewed Journals	Bisserier M, Brojakowska A, Saffran N, Rai AK, Lee B, Coleman M, Sebastian A, Evans A, Mills PJ, Addya S, Arakelyan A, Garikipati VNS, Hadri L, Goukassian DA. "Astronauts plasma-derived exosomes induced aberrant EZH2-mediated H3K27me3 epigenetic regulation of the vitamin D receptor." Front Cardiovasc Med. 2022 Jun 16;9:855181. https://doi.org/10.3389/fcvm.2022.855181 ; PMID: 35783863; PMCID: PMC9243458 , Jun-2022
Articles in Peer-reviewed Journals	Goukassian D, Arakelyan A, Brojakowska A, Bisserier M, Hakobyan S, Hadri L, Rai AK, Evans A, Sebastian A, Truongcao M, Gonzalez C, Bajpai A, Cheng Z, Dubey PK, Addya S, Mills P, Walsh K, Kishore R, Coleman M, Garikipati VNS. "Space flight associated changes in astronauts' plasma-derived small extracellular vesicle microRNA: Biomarker identification." Clin Transl Med. 2022 Jun;12(6):e845. https://doi.org/10.1002/ctm2.845 ; PMID: 35653543; PMCID: PMC9162436 , Jun-2022
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Articles in Peer-reviewed Journals	Bisserier M, Shanmughapriya S, Rai AK, Gonzalez C, Brojakowska A, Garikipati VNS, Madesh M, Mills PJ, Walsh K, Arakelyan A, Kishore R, Hadri L, Goukassian DA. "Cell-free mitochondrial DNA as a potential biomarker for astronauts' health." J Am Heart Assoc. 2021 Nov 2;10(21):e022055. https://doi.org/10.1161/JAHA.121.022055 ; PubMed PMID: 34666498; PMCID: PMC8751818 , Nov-2021